

GRUT — Public Record Update (27 September 2026)

The GR-2 gravitational-forcing campaign: completion, verdict, and dependency boundary
— and the v4 termination status

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Contents

1. The GR-2 campaign is complete and adjudicated	3
2. The scientific verdict	4
3. The dependency boundary	4
4. The v4 termination status (separately adjudicated; quarantined)	5
5. The coupling-class wording (the standing GR2-a editorial instruction, applied)	5
6. The major nonclaims (on the face of this record)	6
7. Provenance and auditability	6
8. What follows	6
9. Addendum (28 September 2026): the termination stop has fired	6

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What this document is. This is the **single dated update** to the program’s public record of 26 September 2026 — the consolidated paper *GRUT — Grand Responsive Universe Theory: Consolidated Theory, Formal Architecture, Results, Limits, and Completion Program* (paper DOI [10.5281/zenodo.22983638](#); concept DOI [10.5281/zenodo.19803663](#); software release `public-record-2026-09-26`). **The published snapshot is not modified**; this update is the new dated source boundary and supersedes the snapshot exactly where they differ, upgrading no claim anywhere. The standing disclosures are unchanged: no external peer review, no experimental validation, and no confirmed prediction is claimed.

Every result below was produced under the program’s method: a charter frozen by commit **before** its instrument ran; a single recorded run; instrument defects disclosed and never repaired silently; red gates preserved under every outcome; a hard stop after each verdict; and a separately recorded owner ruling adjudicating each record at, and never above, its recorded strength.

1. The GR-2 campaign is complete and adjudicated

The campaign (`GR2_CAMPAIGN_DIRECTIVE_01.md`) attacked the gravitational identification layer by layer: does the GRUT generative core *force* the coupling, the spin, universal reach, or the causal cone — or is each supplied? Six frozen-charter forks ran, each adjudicated by owner ruling:

fork	frozen question	verdict	battery
GR2-L6	is the GR-1 3D failure a dimensional inconsistency?	L6-MECHANISM-DERIVED	15/15
GR2-a	does the earned core select the coupling class?	A-COUPLING-NOT-SELECTED	26/26
GR2-b	does the earned core select spin-2?	B-SPIN-NOT-SELECTED	20/20
GR2-c	does anything earned force universal reach ($g_1 = \dots = g_N$)?	C-REACH-NOT-FORCED	16/16
GR2-d	can a universal causal cone emerge from the earned structure?	D-PARTIAL	22/25
GR2-d2	can genuinely different quantum propagation cones be distinguished and realized?	D2-QCONE-DISTINCT	9/9

(Charter commits and artifact hashes: L6 `29ee4d7/64d20c89...`; a `e5687b0/a5f15017...`; b `940ef46/effbb5ab...`; c `c12edfd/452a052b...`; d `ff8db42+Amendment 18624a6/c1718e6c...`; d2 `75f2069/b512cbee...`. The Layer-7 synthesis: `GR2_CAMPAIGN_SYNTHESIS_01.md`, commit `12d896a`.)

GR2-d is recorded **as a partial**: its quantum leg missed a frozen threshold and the failure is preserved, not repaired; GR2-d2 — a narrow, owner-authorized re-charter of that one gate, with analytic pre-run parameter separation, non-target calibration, and blind thresholds — then established the quantum demonstration cleanly.

2. The scientific verdict

GRAVITY_UNDERDETERMINED — from the campaign directive’s pre-registered verdict set {GRAVITY_FORCED, GRAVITY_PARTIALLY_FORCED, GRAVITY_CONDITIONAL, GRAVITY_UNDERDETERMINED, GRAVITY_REFUTED_WITHIN_CLASS}.

The generative core neither forces the gravitational identification nor contradicts it: at every tested coordinate the gravitational constructions remain fully **compatible** and entirely **unselected**. The supplied-side derivations already in the record (U-1’s exchange universality, RS-1’s class selection, TT-1’s channel classification) remain conditionally derived, exactly as recorded; the campaign’s contribution is to have measured the *earned* side of that conditional at four coordinates and found it empty.

Scope, stated on the face of the verdict: this is an adjudication of the tested model classes and adjudicated batteries. It is not a universal mathematical theorem.

3. The dependency boundary

The earned spine stands, within the tested classes:

local ontology $\rightarrow$ continuum / finite reduction $\rightarrow$ memory / irreversible response $\rightarrow$ influence hierarchy ($\mathfrak{A}$).

The four gravitational branches **do not follow** from it:

- $\mathfrak{A} \nRightarrow$ **coupling class (GR2-a)**. The tested five-channel family explicitly contains exponent classes 0, 4, 8, 12, and every earned selector is blind to the distinction. The ω^7 locus is the coefficient condition $c_1 = 0$, $c_2 + c_3 = 0$ — the pair-channel expression of *supplied* conservation/tracelessness structure — nonunique at fixed exponent, and its cancellation is destroyed by an earned-admissible admixture (class 8 $\rightarrow$ class 0).
- $\mathfrak{A} \nRightarrow$ **spin (GR2-b)**. The earned battery eliminates 0 of 6 representation candidates; even the record’s supplied layer (masslessness, Lorentz/gauge structure) leaves scalar, vector, and spin-2 all standing. The classic eliminators — universal attraction (removes the vector) and light bending (removes the Nordström-type scalar) — are empirical inputs found nowhere in the record’s earned or supplied layers.
- $\mathfrak{A} \nRightarrow$ **universal reach (GR2-c)**. The joint influence cone is rank-1 positive for *every* real coupling vector — the earned cone does not force a common coupling sign, let alone a common magnitude; 0 of 14 non-universal assignments were eliminated. The only forcing of $g_A = g_B$ anywhere in the record is the supplied massless gauge structure, acting only on exchanging sectors — and that every sector sits in the exchange-coupled component is itself supplied.
- $\mathfrak{A} \nRightarrow$ **universal causal cone (GR2-d, GR2-d2)**. No earned test eliminates any different-cone system (0/7 classical-and-relabeled, 0/1 quantum); exchange *composes* supplied cones into an exact average rather than selecting one; the earned frequency-domain battery is exactly blind to the cone for every representation class; and an explicitly quantum pair of earned-admissible sectors with analytically separated (1.809 vs 1.000), measurably distinct causal cones is exhibited. GR2-d’s own conjunctive gates V-2/V-3 remain failed at recorded strength — the quantum irreducibility claim is carried by GR2-d2’s record, not by a repair of GR2-d’s.

And the 3D red gate stays red (GR2-L6): the GR-1 3D slope failure is now mechanism-derived (a finite-size/window law, blindly validated at five sizes) — an *explanation* of the red gate, not a pass; no gravitational status was upgraded by it.

The campaign’s graded vocabulary is part of the finding: **derived · derived-in-class · conditionally derived · supplied · compatible · not selected · not forced · demonstrated**

counterexample · open — more informative than a binary “gravity works/doesn’t work.”

The boundary, in one statement: nothing tested in the GRUT generative core forces gravity. The core generates continuum, memory, irreversible response, an influence hierarchy, and spatial geometry recovery — and **{causal cone | coupling | spin | universal reach}** are **additional structural inputs, not consequences**. The supplied gravitational structure cannot retrospectively be relabeled as something generated by the core.

4. The v4 termination status (separately adjudicated; quarantined)

The program’s signed termination condition (provenance/prereg/PREREG_TERMINATION_V4_2026-08-10.txt, sha256 f4bc613c..., in force since its logged signing of 2026-08-10) was reconciled criterion-by-criterion against its own sealed wording and canonical event log (V4_TERMINATION_RECONCILIATION_01.md, adjudicated by owner ruling). **The reconciliation and the GR2 verdict are two independent records: one scientific, one historical/procedural. Neither rewrites the other.**

item	reconciled status
R5(i) — the 2026-12-31 date	not satisfied
R5(ii) — both Part-7 fronts discharged	not satisfied
R5(iii) — unconditional cut-class reply	not satisfied
C1 — the dispatch (pole-vs-cut)	not satisfied (unsent; held)
C2 — DESI DR3	not tested (no release landed; no frozen threshold, so its data outcome presently cannot fire)
C3 — the TT-auto calc	not satisfied (unrun)
C4 — the method (Q3)	not satisfied — 0/3 legs
C5 — the ξ_{ij}/T_T calc	not satisfied (unrun; owed-or-retired)
stop trigger	none fired
the v4 condition	still in force / running
the deposit	not yet due

Three provenance protections from the reconciliation, preserved here:

- (i) *not satisfied* and *not tested* are distinct — C2 is neither a failure nor a success while its own frozen-threshold requirement is unmet; (ii) the “C4” name collision is recorded: GR2’s completion-problem “C4” (irreducibility certificates) is unrelated to v4’s C4 (method-novelty graduation), and GR2’s in-house error catches — the method catching *its own* errors — are precisely what v4’s Q3 excludes and count for nothing there; (iii) the v4 TT fronts presuppose, in their channel’s physics, the per-sector $v = c$ structure that GR2 has now graded as supplied rather than derived — which alters no v4 wording and no v4 status: **GR2 finding $\neq$ v4 criterion modification**. Any future C3 treatment must carry those premises explicitly rather than silently inheriting them as consequences of GRUT.

5. The coupling-class wording (the standing GR2-a editorial instruction, applied)

The 26 September 2026 snapshot reads, in its Class-4 discussion: “*the coupling class is supplied (I1–I6). Class-4 has never been passed.*” Per the owner’s standing instruction recorded at the GR2-a acceptance, this and all future dated records state it at the demonstrated strength:

The coupling class is supplied; within the tested local family it is nonunique at fixed exponent, and the ω^7 cancellation is unstable to an earned-admissible admixture.

No claim is upgraded by this wording. Class-4 has never been passed (unchanged).

6. The major nonclaims (on the face of this record)

- **GR2 does not refute gravity.** Spin-2, universal reach, and a common causal cone remain fully compatible with everything earned.
- **GR2 does not derive gravity.** No earned discriminator selecting any gravitational coordinate exists anywhere in the record.
- **GR2 does not establish a universal causal cone** — neither its presence (not derived) nor its absence (the empirical light-speed/Lorentz constraints stayed outside the earned machinery throughout, by explicit rule, and remain external data).
- **v4 has not reached a termination trigger.** In-house calculation remains permitted under the running clock; the deposit is a future obligation.
- Standing and unchanged: no external peer review; no experimental validation; zero confirmed novel predictions; all red gates preserved, GR-1’s 3D gate included; Class-4 never passed.

7. Provenance and auditability

The campaign’s defect record is part of the scientific record, kept in aggregate and never erased by rerunning until a preferred outcome appeared: two instrument-defect run-1 invalidations caught by halt-grade analytic identity gates (GR2-b’s closure-dimension breach; GR2-c’s eigensolver-scaling breach), one pre-run charter amendment (GR2-d, appended, with its lost blindness disclosed), one preserved gate failure (GR2-d’s Q-2 and its conjunctive V-2/V-3), one thin disclosed margin (GR2-d2’s S-1 at 1.5111 against a floor of 1.5), and one incidental recorded fact (the record’s two mixed-field chain parameter sets have Lieb–Robinson fronts within ~8%). Everything is verifiable from the repository (`ryangrvr/GRUT-RAI`, branch `master-w25bu9`), the Issue #2 adjudication thread, the sha-hashed result artifacts, the sealed v4 file, and its canonical event log.

8. What follows

- The **v4 deposit**, in v4’s own one-line-per-channel form, comes due within one calendar month after the stop fires (earliest candidate: 2026-12-31).
- The two **Part-7 fronts** remain owed-or-retired, exactly as recorded.
- The **dispatch** remains unsent and held, with the logged ill-posedness evidence and a recorded — unadopted — candidate re-pose.
- No **GR2-e** is chartered: with all four prerequisites adjudicated as unforced primitives, “does the core force an Einstein-like action” is no longer well posed as a forcing question; whether an Einstein–Hilbert structure can emerge from some deeper *extension* is a new question for a new charter, not a residue of this one.

This update was adjudicated by owner rulings recorded in the repository (`GR2_SYNTHESIS_OWNER_RULING_01.md`, `V4_RECONCILIATION_OWNER_RULING_01.md`) and is the program’s dated source boundary as of 27 September 2026.

9. Addendum (28 September 2026): the termination stop has fired

Sections 4 and 6 of this update reported, correctly as of its date, that no v4 stop trigger had fired. **On 28 September 2026 the owner adjudicated R5(ii) as SATISFIED** under the sealed condition’s own calc-completed branch — “*both Part-7 fronts discharged (calc completed, or retired with a statement naming what dies and why)*” — because both fronts’ calculations are

now complete: the TT-auto gate's owed Boltzmann-grade common-mode check (A6, discharged 5/5, with every $\kappa=1$ exclusion surviving the correction and the $\kappa=3$ survivals strengthening) and the dissipation-channel closures (the parameter-free Γ_T pinned 62.7 orders below the shared-slot bound, 2026-09-06; the ξ_{ij} stochastic background computed Boltzmann-dead — $e^{(-2\pi\omega/H_0)}$ — at every licensed frequency, parameter-free and kernel-independent, 2026-09-27). **In-house physics calculation is stopped from 28 September 2026; the deposit is due within one calendar month.**

The channel adjudications are unchanged by the stop: C3 remains STILL OPEN (mixed members) — completing its calculation did not satisfy its criterion — and the other channels stand as reconciled. The scientific verdict, GRAVITY_UNDERDETERMINED, is untouched: the two statuses are compatible and independent, exactly as §4 required.

The recorded next scientific direction — the **Level-0 descent**: the split between generative (pre-universe) and emergent (in-universe) rules, and the necessity sweep of the never-axiomatized local-finite substrate — is **design-only under the stop** (LEVEL0_DIRECTION_01.md in the repository). The program's own summary of the moment: the right next question was found at exactly the point where the pre-committed rules require the in-house program to stop asking it — which is what the termination condition was designed to accomplish.